



Landslide volumes and landslide mobilization rates in Umbria, central Italy

Fausto Guzzetti ^{a,*}, Francesca Ardizzone ^a, Mauro Cardinali ^a, Mauro Rossi ^a, Daniela Valigi ^b

^a IRPI CNR, via Madonna Alta 126, 06128 Perugia, Italy

^b Dipartimento di Scienze della Terra, Università di Perugia, Piazza dell'Università, 06123 Perugia, Italy

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ABSTRACT

A catalogue of 677 landslides of the slide type was selected from a global database of geometrical measurements of individual landslides, including landslide area (A_L) and volume (V_L). The measurements were used to establish an empirical relationship to link A_L (in m^2) to V_L (in m^3). The relationship takes the form of a power law with a scaling exponent $\alpha=1.450$, covers eight orders of magnitude of A_L and twelve orders of magnitude of V_L , and is in general agreement with existing relationships published in the literature. The reduced scatter of the experiential data around the dependency line, and the fact that the considered landslides occurred in multiple physiographic and climatic environments and were caused by different triggers, indicate that the relationship between V_L and A_L is largely independent of the physiographical setting. The new relationship was used to determine the volume of individual landslides of the slide type in the Collazzone area, central Italy, a 78.9 km^2 area for which a multi-temporal landslide inventory covering the 69-year period from 1937 to 2005 is available. In the observation period, the total volume of landslide material was $V_{LT}=4.78 \times 10^7 m^3$, corresponding to an average rate of landslide mobilization $\phi_L=8.8 mm yr^{-1}$. Exploiting the temporal information in the landslide inventory, the volume of material produced during different periods by new and reactivated landslides was singled out. The wet period from 1937 to 1941 was recognized as an episode of accelerated landslide production. During this 5-year period, approximately 45% of the total landslide material inventoried in the Collazzone area was produced, corresponding to an average rate of landslide mobilization $\phi_L=54 mm yr^{-1}$, six times higher than the long term rate. The volume of landslide material in an event or period was used as a proxy for the magnitude of the event or period, defined as the logarithm (base 10) of the total landslide volume produced during the event, or period. With this respect, the new relationship to link A_L and V_L is a starting point for the adoption of a quantitative, process based landslide magnitude scale for landslide events.

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1. Introduction

Landslides are caused by various triggers, including earthquakes, rainfall and rapid snowmelt, and are influenced by multiple factors, such as topography, soil and rock types, fractures and bedding planes, and moisture content (Crozier, 1986; Turner and Schuster, 1996). Knowing the number, area, and volume of landslides is important to determine landslide susceptibility and hazard (Soeters and van Westen, 1996; Guzzetti et al., 1999; Malamud et al., 2004a), to ascertain landslide risk (Cardinali et al., 2002; Reichenbach et al., 2005), for forestry, wildlife and ecological studies (Montgomery et al., 2000; Miller and Burnett, 2007), and to evaluate the long-term evolution of landscapes dominated by mass-wasting processes (Hovius et al., 1997; Harmon and Doe, 2001; Lavé and Burbank, 2004; Malamud et al., 2004b; Korup, 2005a,b; Imaizumi and Sidle, 2007; Guzzetti et al., 2008).

The number of landslides in an area is information easily obtained where accurate and reasonably complete landslide inventory maps

are available (Guzzetti et al., 2002; Malamud et al., 2004a; Galli et al., 2008). Where landslide maps are available in digital form, the number of landslide per unit area (i.e., landslide density), the area of individual landslides, and the total landslide area can be calculated. Where multi-temporal landslide inventory maps are available in digital form (Guzzetti et al., 2005, 2006; Imaizumi and Sidle, 2007; Galli et al., 2008), statistics of the number, density and area of landslides can be calculated for different periods.

Determining the volume of a landslide is a more difficult task that requires information on the surface and sub-surface geometry of the slope failure. This information is difficult and expensive to collect. Estimating the volume of slope failures for a large population of landslides (hundreds to several thousand failures) in an area is an even more challenging task (Malamud et al., 2004a) that, at present, can be achieved only by adopting empirical relationships to link the volume of individual landslides to geometrical measurements of the failures, chiefly landslide area (Simonett, 1967; Rice et al., 1969; Innes, 1983; Hovius et al., 1997; Guthrie and Evans, 2004a; Korup, 2005b; ten Brink et al., 2006; Imaizumi and Sidle, 2007; Guzzetti et al., 2008; Imaizumi et al., 2008).

In this paper, we first describe a catalogue of 5814 landslides for which measures of the area, A_L , and volume, V_L , are available. Next, we

* Corresponding author. Tel.: +39 075 5014413; fax: +39 075 5014420.

E-mail address: Fausto.Guzzetti@irpi.cnr.it (F. Guzzetti).

use a subset of this catalogue listing 677 mass movements of the slide type (Cruden and Varnes, 1996) – or predominantly of the slide type – to determine an empirical relationship linking A_L and V_L , and we compare the new relationship to similar relationships in the literature (Simonett, 1967; Rice et al., 1969; Innes, 1983; Guthrie and Evans, 2004a; Korup, 2005b; ten Brink et al., 2006; Imaizumi and Sidle, 2007; Guzzetti et al., 2008; Imaizumi et al., 2008). We exploit the obtained dependency to determine the total volume of landslide material and the rate at which landslides are mobilized in an area of Umbria, central Italy, for which a detailed multi-temporal landslide inventory map is available (Guzzetti et al., 2006; Galli et al., 2008). Lastly, we propose a landslide event magnitude scale based on the total landslide volume produced during an event, or period.

2. Dependency of V_L on A_L

A database of geometrical measurements for individual landslides, including landslide area A_L (in m^2), and volume V_L (in m^3), was compiled through a worldwide literature search, comprising main reference works (e.g., Voight, 1978, 1979; Eisbacher and Clague, 1984; Sassa 1999; Evans and DeGraff, 2002), international journals, conference proceedings, and event and technical reports. First, papers presenting relationships that link geometrical properties of slope failures (i.e., landslide length, width or area) to landslide volume were analysed, including Simonett (1967), Rice et al. (1969), Innes (1983, 1985), Hovius et al. (1997), Guthrie and Evans (2004a), Korup (2005a,b), ten Brink et al. (2006), Imaizumi and Sidle (2007), Guzzetti et al. (2008), and Imaizumi et al. (2008). Next, papers presenting individual measurements of landslide area and volume for multiple slope failures in an area were studied, including Rice et al. (1969), Rice and Foggin (1971), Abele (1974), Whitehouse (1983), Larsen and Torres Sanchez (1998), Martin et al. (2002), and Hafliadason et al. (2005). Lastly, reports describing single landslides for which the geometrical characteristics were measured, estimated, or inferred from surface and subsurface investigations were considered.

The collected information was organized in a database listing 5654 terrestrial (97.2%) and 160 subaqueous (2.8%) mass movements. For each landslide, the information included: (i) the geographical location of the slope failure, (ii) the landslide geometrical properties, including length, width, depth, area, and volume (Cruden and Varnes, 1996), (iii) the predominant type of movement (Cruden and Varnes, 1996), and (iv) the main rock type. Not all the information was available for all the landslides. For some of the landslides the geometrical properties were known accurately. For other slope failures a range of values was given (e.g., Sassa, 1996; Plaza-Nieto et al., 1999; Schuster et al., 2002; Chang et al., 2005). In the latter case, the average value was considered. For some of the landslides, the geometrical properties were inferred from maps and aerial photographs (i.e., length, width, area) and from cross sections (depth).

Geomorphological considerations suggest that measurement of V_L in the field, from aerial photographs or topographic maps, is sufficiently accurate for the purpose of our work. An example will prove the concept. Fig. 1 shows a deep-seated slide in Umbria, central Italy. The length of the slide is approximately 400 m and the width at the toe is around 120 m, i.e. $A_L \approx 4.8 \times 10^4 m^2$. Considering that the landslide depth D_L in the source area is between 12 m and 25 m, the volume of the displaced material is in the range $6 \times 10^5 m^3 \leq V_L \leq 1.2 \times 10^6 m^3$, and most probably $V_L = 9 \times 10^5 m^3$. Hence, the uncertainty in the measurement of the volume is within the same order of magnitude of the computed volume. Similar considerations can be made for most landslides.

From the available database of terrestrial landslides, 677 (12.0%) slope failures of the slide type (Cruden and Varnes, 1996) – or predominantly of the slide type – were singled out, including 207 landslides inventoried by Simonett and Schuman in New Guinea (Simonett, 1967), 66 soil slips occurred in the San Dimas Experimental

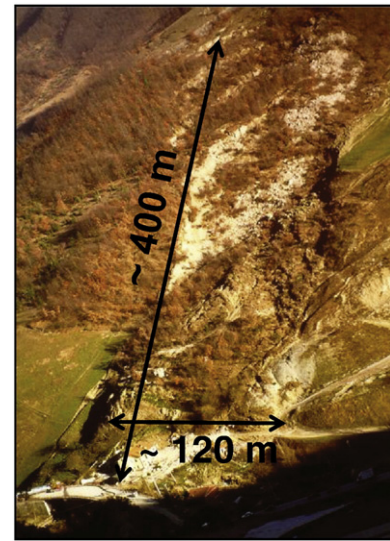


Fig. 1. Valderchia landslide, Umbria, central Italy. The slope failure of the slide type was triggered by rapid snow melting on 6 January 1997.

Forest in southern California in the period from 1966 to 1969 (Rice et al., 1969; Rice and Foggin, 1971), 28 large landslides in China (Wen et al., 2004), 16 landslides measured in south-eastern Norway (Jaedicke and Kleven, 2007), and 14 landslides mapped along the Cromwell Gorge, in the Otago region of New Zealand's South Island (Gillon and Hancox, 1992). Soil slips, soil slides, and debris slides that evolved into debris flows (e.g., Innes, 1983; Martin et al., 2002) were excluded where a clear distinction between the geometry of the source, transport and deposition areas was not possible. Submarine landslides (e.g., ten Brink et al., 2006) were also excluded, because the mechanics of subaqueous slope failures is considered different from that of terrestrial landslides (Hafliadason et al., 2005; ten Brink et al., 2006).

The 677 landslides for which information on A_L and V_L was available were plotted in a single graph, in log–log coordinates (Fig. 2). In the graph, A_L (x-axis) covers 8 orders of magnitude ($2.1 \times 10^0 m^2 \leq A_L \leq 7.0 \times 10^7 m^2$), V_L (y-axis) spans 12 orders of magnitude ($3.4 \times 10^{-1} m^3 \leq V_L \leq 2.9 \times 10^{10} m^3$), and the accuracy of each landslide measurement depends on the size of the landslide. Visual inspection of Fig. 2 reveals a distinct linear (in log–log coordinates) relationship between A_L and V_L for multiple orders of magnitude. This suggests a self-similar behavior of the dependency between landslide area and volume. The observation is important because it allows using the section of the catalogue where the information is most abundant (i.e., $6 \times 10^1 m^2 < A_L < 2 \times 10^6 m^2$) to infer a dependency for the sections of the catalogue where the information is less numerous, e.g. for very small and for very large landslides. Further inspection of Fig. 2 reveals that the scatter of the empirical data around the central tendency line is limited, even considering that the data are shown in log–log coordinates. This is remarkable, given the uncertainty associated with the measurement of A_L and V_L , and the fact that the shown landslides occurred in different lithological, morphological and climatic settings, and were triggered by different causes, including rainfall, earthquakes, and rapid snowmelt.

To model the empirical relationship between A_L and V_L , an equation of the form $V_L = \varepsilon \times A_L^\alpha$ was fitted to the empirical data. To account for problems associated with the fitting of data spanning multiple orders of magnitude (e.g., the least square minimization criteria may not work), the empirical data were log-transformed. Different fitting techniques were tested on the log-transformed data, including least square linear fit (Chambers, 1992; Wilkinson and Rogers, 1973), robust linear fit (Marazzi, 1993; Venables and Ripley, 2002), robust resistant regression (Marazzi, 1993), and least square non-linear fit (Bates and Chambers, 1992). Results of the application of the different techniques were very similar, with $0.070 \leq \varepsilon \leq 0.087$,

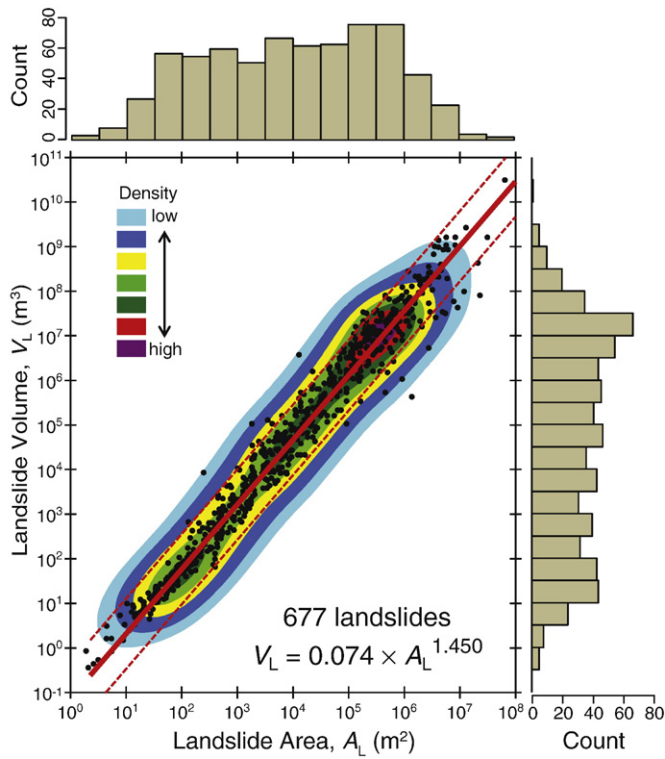


Fig. 2. Empirical measurements for landslides of the slide type obtained through a literature search. Dots portray the area, A_L (x-axis, m^2), and volume, V_L (y-axis, m^3), of 677 landslides. Thick red line is best fit obtained adopting a robust linear fitting technique. Dashed red lines show 95% confidence intervals. Colours indicate density of points obtained through bivariate kernel density estimation. Top and right histograms show count of A_L and V_L , respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

$1.429 \leq \alpha \leq 1.452$, and $0.9707 \leq R^2 \leq 0.9709$. In an attempt to minimize the effects of the outliers, the result obtained through robust linear fitting (Venables and Ripley, 2002) was adopted. This resulted in the relationship:

$$V_L = 0.074 \times A_L^{1.450} \quad (R^2 = 0.9707) \quad (1)$$

with A_L in m^2 , V_L in m^3 , and a standard error of the scaling exponent $\alpha = 0.0086$. This power law equation can be used to estimate the

volume of individual landslides of the slide type when the area of the slope failure is known.

3. Comparison with existing relationships

Other relationships linking A_L to V_L are available in the literature (Table 1 and Fig. 3). Simonett (1967), working in the Bewani and Torricelli Mountains in central New Guinea, estimated in the field the area and volume of 207 landslides, and obtained the relationship $V_L = 0.2049 \times A_L^{1.368}$, for $2.5 \times 10^1 \text{ ft}^2 \leq A_L \leq 2 \times 10^6 \text{ ft}^2$. Rice et al. (1969) measured the length, width, area and volume of 29 soil slips in southern California, and determined the relationship $V_L = 0.234 \times A_L^{1.11}$, for $2.1 \times 10^0 \text{ m}^2 \leq A_L \leq 2 \times 10^2 \text{ m}^2$. Innes (1983) estimated the width, length and volume of 30 debris flow deposits in the Scottish Highlands, and obtained the relationship $V_L = 0.0329 \times A_L^{1.3852}$, for $3 \times 10^1 \text{ m}^2 \leq A_L \leq 5 \times 10^2 \text{ m}^2$. Landslide area was calculated as the product of landslide width and length. Guthrie and Evans (2004a) studied 124 debris slides in the west coast of Vancouver Island, British Columbia, and found $V_L = 0.1549 \times A_L^{1.0905}$, for $7 \times 10^2 \text{ m}^2 < A_L < 1.2 \times 10^5 \text{ m}^2$. Korup (2005b) studied 23 landslides with $A_L > 1.2 \times 10^6 \text{ m}^2$ in the Western Southern Alps, New Zealand, and established that for these large landslides $V_L = 0.02 \times A_L^{1.95}$, with A_L in km^2 and V_L in km^3 . Imaizumi and Sidle (2007) measured the volume of 51 shallow landslide scars in the Miyagawa catchment, central Japan, and found that $V_L = 0.39 \times A_L^{1.31}$, for $1 \times 10^1 \text{ m}^2 < A_L < 3 \times 10^3 \text{ m}^2$. Imaizumi et al. (2008), also working in Japan, used measurements of 11 landslides to determine that $V_L = 0.19 \times A_L^{1.19}$, for $5 \times 10^1 \text{ m}^2 < A_L < 4 \times 10^3 \text{ m}^2$. Guzzetti et al. (2008), using a preliminary version of the database used in this work listing 539 landslides worldwide, obtained the relationship $V_L = 0.0844 \times A_L^{1.4324}$, for $1 \times 10^1 \text{ m}^2 < A_L < 1 \times 10^9 \text{ m}^2$.

A few authors presented, or made available to us, information on the geometry of populations of individual landslides from which area-to-volume dependencies could be calculated (Table 1). Rice and Foggin (1971), working in southern California, measured the area and volume of 37 soil slips. Using this dataset, we obtained the relationship $V_L = 0.328 \times A_L^{1.104}$, for $1.1 \times 10^1 \text{ m}^2 < A_L < 1.5 \times 10^3 \text{ m}^2$. Abele (1974) studied different types of landslides in the Alps. For 53 landslide deposits measurements of A_L and V_L were available. Exploiting this dataset, we obtained the relationship $V_L = 1.55 \times A_L^{1.183}$, for $6.0 \times 10^3 \text{ m}^2 < A_L < 6.0 \times 10^7 \text{ m}^2$. Whitehouse (1983) measured 46 large rock avalanche deposits in the Central Southern Alps, New Zealand. Analysis of this dataset revealed an area-to-volume dependency $V_L = 0.769 \times A_L^{1.250}$, for $5 \times 10^4 \text{ m}^2 < A_L < 3.87 \times 10^6 \text{ m}^2$. Larsen and Torres Sanchez (1998) studied shallow landslides and debris

Table 1
Empirical relationships linking landslide area A_L to landslide volume V_L shown in Fig. 3

ID	Equation	Min A_L (m^2)	Max A_L (m^2)	N	Equation	Min A_L	Max A_L	A_L	Source
1	$V_L = 0.074 \times A_L^{1.450}$	2×10^0	1×10^9	677					This work (Eq. (1))
2	$V_L = 0.1479 \times A_L^{1.368}$	2.3×10^0	1.9×10^5	207	$V_L = 0.2049 \times A_L^{1.368}$	2.5×10^1	2.0×10^6	ft ²	Simonett (1967)
3	$V_L = 0.234 \times A_L^{1.11}$	2.1×10^0	2×10^2	29					Rice et al. (1969)
4	$V_L = 0.0329 \times A_L^{1.3852}$	3×10^1	5×10^2	30					Innes (1983)
5	$V_L = 0.1549 \times A_L^{1.0905}$	7×10^2	1.2×10^5	124					Guthrie and Evans (2004a)
6	$V_L = 0.00004 \times A_L^{1.95}$	$> 1 \times 10^6$		23	$V_L = 0.02 \times A_L^{1.95}$	> 1		km ²	Korup (2005b)
7	$V_L = 4.655 \times A_L^{1.292}$	5×10^5	2×10^8	160	$V_L = 0.263 \times A_L^{1.292}$	5×10^{-1}	2×10^2	km ²	ten Brink et al. (2006)
8	$V_L = 0.39 \times A_L^{1.31}$	1×10^1	3×10^3	51					Imaizumi and Sidle (2007)
9	$V_L = 0.0844 \times A_L^{1.4324}$	1×10^1	1×10^9	539					Guzzetti et al. (2008)
10	$V_L = 0.19 \times A_L^{1.19}$	5×10^1	4×10^3	11					Imaizumi et al. (2008)
11(*)	$V_L = 0.328 \times A_L^{1.104}$	1.1×10^1	1.5×10^3	37					Rice and Foggin (1971)
12(*)	$V_L = 0.242 \times A_L^{1.307}$	2×10^5	6×10^7	53					Abele (1974)
13(*)	$V_L = 0.769 \times A_L^{1.250}$	5×10^4	3.9×10^6	45					Whitehouse (1983)
14(*)	$V_L = 1.826 \times A_L^{0.898}$	5×10^1	1.6×10^4	1019					Larsen and Torres Sanchez (1998)
15(*)	$V_L = 1.0359 \times A_L^{0.880}$	2×10^2	5.2×10^4	615					Martin et al. (2002)
16(*)	$V_L = 12.273 \times A_L^{1.047}$	3×10^5	3.9×10^{10}	65					Hafliadason et al. (2005)

Column 1 lists the equation number; asterisks outline relationships computed using measurements available in the literature or made available by the authors. Column 2 shows the equations (A_L in m^2 , V_L in m^3). Columns 3 and 4 give the minimum and maximum values for A_L . Column 5 lists the number of data. Columns 6 to 9 list equations and ranges of application for A_L originally given in different units of measure. Column 10 gives the source of information.

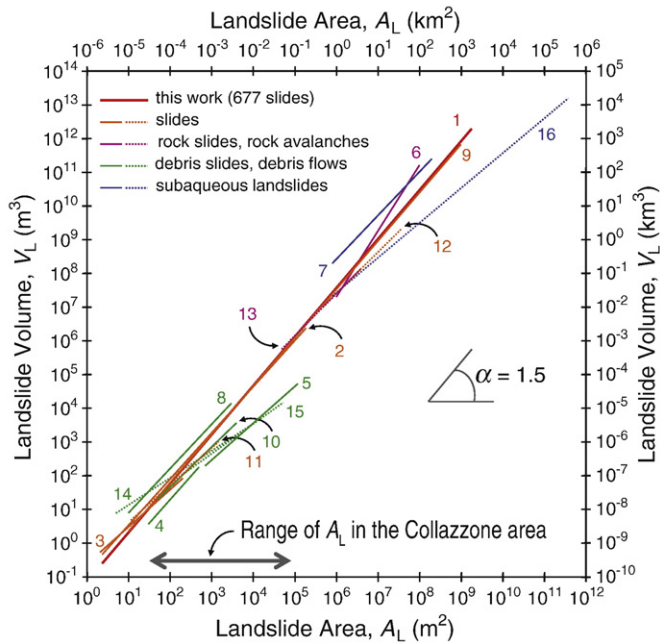


Fig. 3. Empirical relationships proposed in the literature to link landslide area A_L (x-axis) to landslide volume V_L (y-axis) (Table 1). Thick red line (# 1) is the dependency established in this work (Eq. (1)). Colours show relationships for different types of landslides. Continuous lines are dependencies proposed by authors. Dotted lines are dependencies obtained from empirical data published in the literature or made available to us. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

flows in three study areas in Puerto Rico. Landslides were identified on stereoscopic aerial photographs, and the area of the individual landslides was calculated as length $\times$ width, assuming a rectangular shape for the slope failure. Similarly, landslide volume was obtained multiplying landslide area by the average soil depth, ascertained in the field at few localities. Using this dataset, we obtained the relationship $V_L = 1.826 \times A_L^{0.898}$, for $\approx 5.0 \times 10^1 \text{ m}^2 < A_L < \approx 1.58 \times 10^4 \text{ m}^2$. Martin et al. (2002), in a study of landslides in the Queen Charlotte Islands, British Columbia, measured in the field the area and volume of 45 shallow landslides, chiefly debris slides and debris flows. Using this dataset, we obtained the relationship $V_L = 1.0359 \times A_L^{0.880}$, for $\approx 2.0 \times 10^2 \text{ m}^2 < A_L < \approx 5.18 \times 10^4 \text{ m}^2$.

Fig. 3 shows empirical relationships linking A_L to V_L published in the literature – or obtained from geometrical data published in the literature – together with the new relationship determined in this study (Eq. (1)). Most of the relationships exhibit a similar trend, despite the scatter attributed to different types of slope failures, different criteria adopted to estimate landslide volume, and different climatic, geological and physiographic settings. Where A_L was calculated as the product of landslide width and length (e.g. Innes, 1983; Larsen and Torres Sanchez, 1998), or V_L was obtained as the product of A_L and the average landslide depth D_L (e.g. Larsen and Torres Sanchez, 1998; Martin et al., 2002) the volume of the landslides was probably overestimated. Where the size of the landslide deposit was measured (e.g. Innes, 1983), the volume of the individual landslides was also overestimated; whereas where the size of the landslide scar was measured (e.g. Imaizumi and Sidle, 2007; Imaizumi et al., 2008), the volume of the landslides was probably underestimated.

Inspection of Fig. 3 reveals that Eq. (1) is in reasonably good agreement with most of the published relationships, for multiple orders of magnitude of A_L . This is important, as it suggests that the relationship between the volume and the area of a landslide (and particularly of slope failures of the slide type) is essentially geometrical, and largely independent of the local physiographical setting. Where a weak zone exists at a certain depth in a slope (Katz and Aharonov, 2006) (e.g., the

contact between soil and bedrock, a weak sedimentary layer, a joint), the size of the landslide is determined, although the shape of the landslide depends on local conditions. Close inspection of Fig. 3 and Table 1 reveals that some of the empirical relationships obtained from datasets listing predominantly small landslides (e.g., Rice et al., 1969; Larsen and Torres Sanchez, 1998; Martin et al., 2002) have a smaller scaling exponent than the relationships obtained for large landslides (e.g., Korup, 2005b). This may be a problem of sample size, the result of the method adopted to measure V_L or to obtain the scaling relationship, or it may reflect a change in the scaling of the dependency of V_L from A_L with increasing landslide size.

Finally, we note that in Eq. (1) $\alpha < 1.5$, a scaling exponent for which the shape of the landslide would (on average) be completely self-similar, e.g., $V_L \propto A_L^{1.5}$ and landslide thickness $D_L \propto A_L^{0.5}$. Although our analysis indicates that the scaling is different from 1.5 ($\alpha = 1.449$, standard error = 0.0086), given the relatively small number of data used to determine the relationship (677) and the uncertainties associated to the measurement of landslide size (A_L , V_L), we cannot rule out the possibility of a simple scaling.

4. Application to the Collazzone area

For the Collazzone area, Umbria, central Italy (Fig. 4), Eq. (1) was used to calculate the volume of individual landslides from their (planimetric) area. The volume information was then used to (i) evaluate the total volume of landslide material V_{LT} , (ii) estimate landslide mobilization rates φ_L , and (iii) determine the magnitude of individual landslide events or periods, m_L .

4.1. Study area

The Collazzone area extends for 78.9 km² in central Umbria (Fig. 4A). Elevation in the area ranges from 145 m to 634 m above sea level and the slope, computed from a 10 m $\times$ 10 m DTM, ranges from 0° to 63.7° with a mean value of 9.9°. In the area, the terrain is hilly, valleys are asymmetrical, and the lithology and attitude of bedding control the slopes. Sedimentary rocks Lias to Holocene in age crop out in the area. Soils range in thickness from a few decimetres to more than 1 m; they have a fine or medium texture, and exhibit a xenic moisture regime. Precipitation is most abundant in October and November; with a mean annual rainfall in the period from 1921 to 2001 of 884 mm at the Casalina rain gauge (Fig. 4C). Snow falls on the area on average every 2–3 years. Landslides are abundant in the area (Fig. 4B), and range in age, type, morphology, and volume from very old – partly eroded – large and deep-seated slides to young, mostly shallow slides involving the soil mantle. Slope failures are triggered chiefly by meteorological events, including intense and prolonged rainfall and rapid snowmelt (Cardinali et al., 2000; Ardizzone et al., 2007).

4.2. Multi-temporal landslide inventory map

For the Collazzone area, a detailed multi-temporal landslide inventory map is available (Fig. 4B) (Guzzetti et al., 2006; Galli et al., 2008). The inventory was prepared at 1:10,000 scale through the interpretation of five sets of aerial photographs taken in the period 1941–1997 at scales ranging from 1:13,000 to 1:33,000. The landslide map was updated in the period from 1999 to December 2005 through field surveys conducted following periods of prolonged rainfall to map the rainfall-induced landslides. In the digital geographical database, at 1:10,000 scale, landslides attributed to a single date (e.g., a rainfall event, a rapid snowmelt event) or period were stored separately. Following this procedure, new and active landslides recognized, e.g. in the 1977 aerial photographs were stored in a separate layer than the landslides mapped as inactive in the same photographs. As a result, for a single set of aerial photographs, multiple layers were obtained. The obtained geographical database stores information on landslides

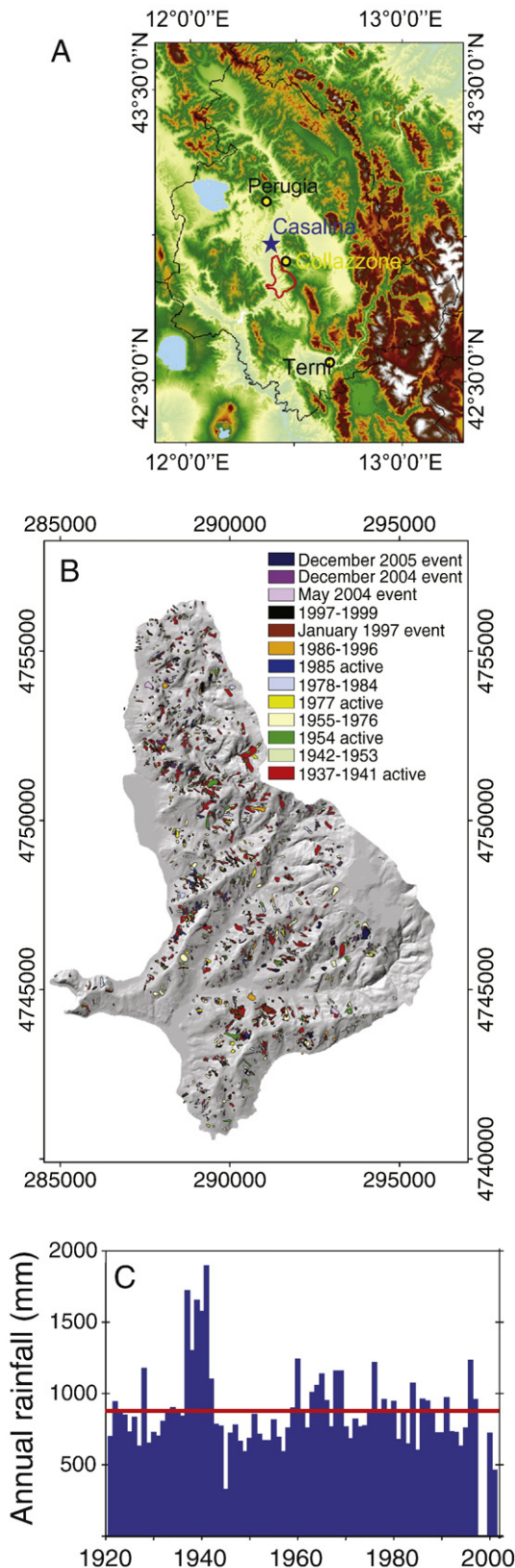


Fig. 4. Collazzone area. (A) Location map. Star is the Casalina rain gauge. (B) Morphology and multi-temporal landslide inventory map. Colours portray landslides of different periods, from 1937 to 2005. (C) Mean annual precipitation (MAP) for the Casalina rain gauge. Red line shows MAP for the 81-year period from 1921 to 2001. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

attributed to 13 dates or periods. The combination of the different layers represents the multi-temporal landslide inventory map for the Collazzone area (Fig. 4B).

The multi-temporal inventory shows 2840 landslides, for a total mapped landslide area of $22.78 \times 10^6 \text{ m}^2$, 21.2% of the territory. Considering only the recent landslides (i.e., the landslides in the period from about 1937 to December 2005), the inventory lists 2543 landslides, for a total mapped landslide area $A_{LT} = 10.43 \times 10^6 \text{ m}^2$, corresponding to a density of 32 slope failures per square kilometre. Due to geographical overlap of landslides of different ages (periods), the total area affected by landslides is $7.86 \times 10^6 \text{ m}^2$, 10.0% of the territory. This is a lower estimate, as the multi-temporal inventory is incomplete for small landslides (Malamud et al., 2004a). Mapped landslides extend in size from $3.59 \times 10^1 \text{ m}^2$ to $7.53 \times 10^4 \text{ m}^2$, and the most abundant failures shown in the map have an area of about $8.15 \times 10^2 \text{ m}^2$ (Guzzetti et al., 2006; Galli et al., 2008).

4.3. Temporal landslide information

Based on the multi-temporal inventory, three types of landslides were singled out, namely: (i) event landslides, (ii) active landslides, and (iii) other landslides. Event landslides were triggered by a known trigger (e.g., a rapid snowmelt event (Cardinali et al., 2000), a rainfall event (Ardizzone et al., 2007)) and were identified and mapped using a single set of aerial photographs taken shortly after the event, or during a single field campaign. Event landslides pertaining to a single temporal layer of the multi-temporal inventory are of the same age, inferred from the date of the event, or the date of the aerial photographs or the field campaign. Active landslides comprise slope failures recognized as active (i.e., fresh) in an individual set of aerial photographs, but that could not be attributed to a specific trigger. Due to their fresh morphological appearance, the inference is made that the landslides formed shortly before the date of the aerial photographs (i.e., during the same wet period). Active landslides in a single layer of the multi-temporal inventory were caused by a single, undetermined trigger (i.e., a rainstorm, a snowmelt event) or a set of triggers (i.e., a wet season, or multiple wet seasons encompassing a number of storms), and were considered to be all of about the same age, inferred from the date of the aerial photographs. Other landslides encompass slope failures identified through the systematic comparison of two subsequent sets of aerial photographs of different vintages. Individual or multiple unknown triggers produced these landslides during the considered period. The exact or approximate age of the landslides remains undetermined, but it is inferred to be intermediate between the dates of the aerial photographs used to identify the landslides.

4.4. Landslide volume

Eq. (1) was used to calculate the volume of the individual landslides based on their area shown in each temporal layer to compose the multi-temporal landslide inventory map (Fig. 4B). Results are summarized in Fig. 5. In the 69-year period from 1937 to 2005, the mapped landslides range in volume from $1.3 \times 10^1 \text{ m}^3$ to $8.7 \times 10^5 \text{ m}^3$, with the most numerous failures in the range $5 \times 10^1 \text{ m}^3 < V_L < 3 \times 10^2 \text{ m}^3$. The probability density of V_L (Fig. 5A) can be compared with the similar probability density for A_L determined by Guzzetti et al. (2006) and Galli et al. (2008) for the same area. In the investigated period, the seven largest landslides (0.3% of total number) account for 10% of the total landslide volume (and 4% of total landslide area), and the 110 largest landslides (5.5% of total number) account for 50% of the total landslide volume (and 32% of total landslide area) (Fig. 5B). The figures confirm the importance of large landslides in determining the total volume of landslide material in an area (Hovius et al., 1997; Guzzetti et al., 2008).

In the studied period, the landslide volume totalled $V_{LT} = 4.78 \times 10^7 \text{ m}^3$, corresponding to an average thickness of mobilized landslide

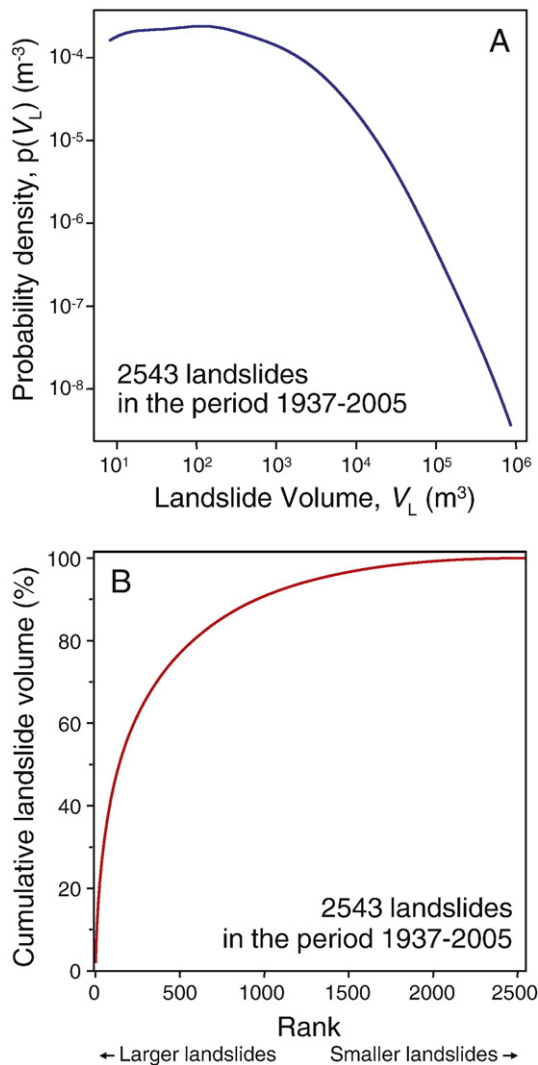


Fig. 5. Collazzone area. (A) Probability density of landslide volume obtained through kernel density estimation. (B) Cumulative proportion of landslide volume versus landslide rank.

material for the study area ($7.89 \times 10^7 \text{ m}^2$) of $\approx 0.46 \text{ m}$. The figure compares to $\approx 5.6 \text{ m}$ obtained by Guzzetti et al. (2008) for the Upper Tiber River basin, central Italy. The difference is significant, and is attributed to the lack of occurrence of very large landslides in the Collazzone area during the considered period. Large landslides ($A_L > 1 \times 10^5 \text{ m}^2$) exist in the area, but are older or much older than 1937 (Guzzetti et al., 2006; Galli et al., 2008) (Fig. 4B). The estimated depth of the individual landslides – calculated as the ratio between the measured A_L and the computed V_L – ranges from 0.4 m to 11.5 m, with an average of 2.7 m and a modal value of 3.6 m.

For each of the considered periods in the multi-temporal inventory (corresponding to a set of event landslides of known age, to a set of active landslides of approximately the same age, or to an ensemble of slope failures of unknown age in an identified period), the total volume of landslide material, including new and reactivated landslides, was computed by summing the volume of the individual slope failures. Landslides of the same (known or inferred) age nested inside larger landslides were excluded from the summation, to avoid counting the same landslide material multiple times. Fig. 6A shows that the produced landslide material varied in the considered period. The majority of the landslide volume ($2.13 \times 10^7 \text{ m}^3$, 44.5% of V_{LT}) was produced during the 1937–41 period. This corresponds to a yearly average for the 5-year period of $4.26 \times 10^6 \text{ m}^3 \text{ yr}^{-1}$, a (average) value

larger than the total volume measured for any of the other considered periods. Analysis of the historical record of daily rainfall for the Casalina rain gauge (Fig. 4C) indicates that the period between August 1937 and July 1941 was the wettest on record, with a mean annual precipitation, MAP = 1636 mm, exceeding by 100% the long-term MAP (884 mm) computed for the 81-year period from 1921 to 2001.

The majority of the landslide volume for the investigated period ($3.53 \times 10^7 \text{ m}^3$, 73.8%) was mobilized by active landslides ($2.91 \times 10^7 \text{ m}^3$, 60.1%) and by event landslides ($6.21 \times 10^6 \text{ m}^3$, 13.0%) (Fig. 6A). Other landslides accounted for $1.25 \times 10^7 \text{ m}^3$, 26.2% of the total landslide volume in the period. Excluding the 1937–41 period of amplified landslide activity, the proportion of landslide volume mobilized by event ($6.21 \times 10^6 \text{ m}^3$) and active ($7.80 \times 10^6 \text{ m}^3$) landslides reduces to $1.40 \times 10^7 \text{ m}^3$, 52.8% of the total landslide volume mapped in the 64-year period from 1942 to 2005. The remaining landslide volume, $1.25 \times 10^7 \text{ m}^3$ (47.2%), was produced by landslides of unknown age during inter-event periods.

Fig. 6A also shows the proportion of the total volume of reactivated and new landslides from 1942 to 2005. During inter-events (green bars), the proportion of landslide volume mobilized by new failures was slightly higher (average = 53.3%) than the proportion mobilized by reactivations (average = 46.7%). Conversely, for active periods (yellow-orange bars) and individual events (red bars) the proportion of

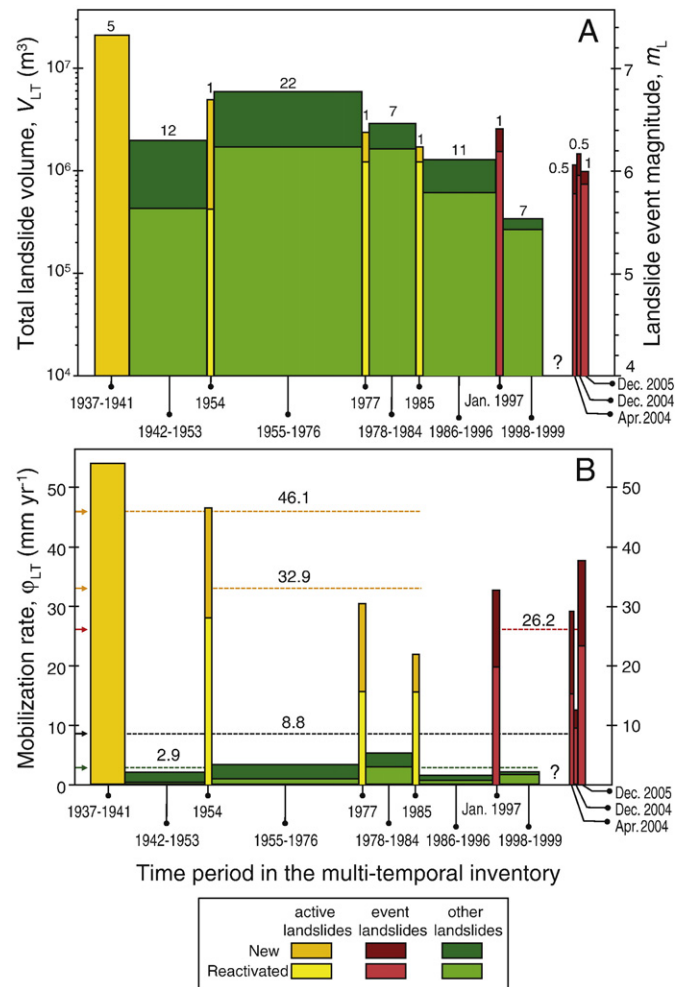


Fig. 6. Collazzone area. (A) Total landslide volume V_{LT} (left y-axis) and corresponding landslide magnitude m_L (right y-axis) for different ages, in the 69-year period from 1937 to 2005. (B) Landslide mobilization rates ϕ_L for different ages, in the same period. Dotted lines show average values for the observation period (black), for event landslides (red), for active landslide periods (orange), and for other landslides (green). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

landslide volume mobilized by new failures was lower (average = 38.0%) than the proportion of material mobilized by reactivations (average = 62.0%). We take this as an indication that during individual events or active periods the landscapes behave differently than during the inter-events.

4.5. Landslide event magnitude

Investigators do not agree on a single measure for landslide magnitude, or intensity (Hungr, 1997; Cardinali et al., 2002; Malamud et al., 2004a; Reichenbach et al., 2005). Malamud et al. (2004a) proposed using the total number of the triggered landslides, information available from accurate event inventory maps. Here, a similar method was adopted, and the total volume of landslides produced during an event (or a period) was used to measure the magnitude of a landslide event (or period). More precisely, the magnitude of a landslide event (or period) m_L was defined as the logarithm (base 10) of the total landslide volume produced during the event (or period). Thus, an event that resulted in $1 \times 10^5 \text{ m}^3$ of landslide material is a magnitude 5 event, and a period during which $1 \times 10^6 \text{ m}^3$ of landslides was formed is a magnitude 6 period. Using this landslide magnitude scale, the period of augmented landslide activity between 1937 and 1941 was $m_L = 7.3$, the January 1997 rapid snowmelt event was $m_L = 6.4$, and the December 2005 rainfall event has $m_L = 6.0$ (Fig. 6A).

4.6. Landslide mobilization rates

Landslides in the Collazzone area are rapid to very slow moving failures chiefly of the slide type (Cruden and Varnes, 1996). After failure, these landslides move down slope for a relatively short distance (from a few metres to a few tens of metres). The displaced material is deposited on or at the bottom of the slope, and is not completely removed (eroded) from the failed slope. For this reason, in the following we refer to landslide mobilization rates ϕ_L , and not to landslide erosion or denudation rates (e.g., Hovius et al., 1997; Martin et al., 2002; Guthrie and Evans, 2004a,b; Korup, 2005a,b; Imaizumi and Sidle, 2007).

Landslide mobilization rates ϕ_L were calculated by dividing the total volume of landslide material in a period (in m^3) by the length of the period (in years) and by the extent of the study area ($7.89 \times 10^7 \text{ m}^2$), and are given in mm yr^{-1} (Fig. 6B). For individual events in a year (i.e., Jan. 1997, Dec. 2005), one year was taken as the length for the period. For multiple events in a year (i.e., Apr. 2004, Dec. 2004), six months were taken as the length for the individual periods. For the 69-year period from 1937 to 2005 the landslide volume totalled $4.78 \times 10^7 \text{ m}^3$, which is an average of $6.93 \times 10^5 \text{ m}^3 \text{ yr}^{-1}$, or a rate $\phi_L = 8.8 \text{ mm yr}^{-1}$. Inspection of Fig. 6B indicates that the mobilization rates varied significantly in the observation period. The largest rate, $4.3 \times 10^6 \text{ m}^3 \text{ yr}^{-1}$ or $\phi_L = 54 \text{ mm yr}^{-1}$, was observed in the 1937–1941 wet period of augmented landsliding, and exceeded by more than 610% the long term (69-year) average. Excluding the 1937–1941 period, the landslide volume totalled $2.65 \times 10^7 \text{ m}^3$ (55.4%), corresponding to $4.1 \times 10^5 \text{ m}^3 \text{ yr}^{-1}$ or a landslide mobilization rate $\phi_L = 5.3 \text{ mm yr}^{-1}$.

Even excluding the 1937–1941 period, most of the landslide material was mobilized during individual landslide events or landslide active periods (Fig. 6B). The average mobilization rates for individual landslide events and for active landslide periods are 9 and 11 times higher than the average rates for the inter-event periods. The fact that the average rates for active landslide periods are higher than the average rates for individual landslide events may confirm that slope failures in active periods were triggered by individual events that went unreported. Inspection of the precipitation record (Fig. 4C) indicates that rainfall events occurred in the observation period. We conclude that landslides in the Collazzone area are mobilized primarily by intense or prolonged meteorological events. The finding that, in the investigated period, the average landslide depth D_L in the Collazzone area is about one order of magnitude smaller than the

average D_L for the Upper Tiber River basin (Guzzetti et al., 2008), in an undefined by much longer period (≈ 0.47 vs. $\approx 5.6 \text{ m}$), is an indication that in Umbria landslide mobilization rates vary significantly, in time and geographically.

4.7. Accuracy of the measurements

The accuracy of the measurements of V_L obtained for the individual landslides depends on the accuracy of the measures of A_L and the reliability of Eq. (1). Accuracy of the measurements of the total landslide volume V_{LT} for the entire observation period and for individual events or periods, depends on the quality of the individual volume measurements and the completeness of the landslide inventory. The latter depends on the type of temporal layer and the landslide size. Layers showing event landslides (red bars in Fig. 6) are considered complete, for all landslide sizes (Malamud et al., 2004a). Layers of active landslides (yellow bars) are considered complete for the large and the medium size landslides, and probably incomplete for small and very small landslides. Temporal layers covering inter-periods (green bars) are incomplete, and their degree of completeness is unknown.

Because some of the inventory layers are incomplete, the obtained figures for V_{LT} are lower estimates. This reflects on the estimates of landslide magnitude m_L and on the mobilization rates ϕ_L , which are also lower values. However, inspection of Fig. 5B reveals that V_{LT} depends on the (rare) large failures, and is not influenced significantly by the (numerous) small failures. We conclude that, despite incompleteness, measures of V_{LT} and estimates of m_L and ϕ_L are reasonably accurate.

5. Conclusions

An empirical relationship to link landslide area (A_L in m^2) to landslide volume (V_L in m^3) was obtained from a worldwide catalogue of 677 landslides of the slide type. The relationship is a power law with a scaling exponent $\alpha = 1.450$, and is in general agreement with similar relationships in the literature (Simonett, 1967; Rice et al., 1969; Innes, 1983; Guthrie and Evans, 2004a; Korup, 2005b; ten Brink et al., 2006; Imaizumi and Sidle, 2007; Guzzetti et al., 2008; Imaizumi et al., 2008). We suggest that the relationship is largely geometrical, and not influenced significantly by geomorphological or mechanical properties of the failed soils or rocks, or the landslide types.

The empirical relationship was applied to a multi-temporal landslide inventory for the Collazzone area, central Italy, covering unsystematically the period 1937–2005 (Guzzetti et al., 2006; Galli et al., 2008). Inspection of the landslide and precipitation records indicates that landslides were mobilized primarily by intense or prolonged rainfall events and by snowmelt events. The total volume of landslide material in the 69-year observation period was ascertained $V_{LT} = 4.78 \times 10^7 \text{ m}^3$, corresponding to an average mobilization rate $\phi_L = 8.8 \text{ mm yr}^{-1}$. Exploiting the chronological information available in the multi-temporal inventory, slope failures were classified as event landslides, active landslides and other landslides, and the volume of landslide material mobilized during the different periods by reactivations and new landslides was determined. Results indicate that the period from 1937 to 1941 – the wettest on record – was an episode of accelerated landslide production. During this 5-year period, approximately 45% of the total inventoried landslide material ($2.1 \times 10^7 \text{ m}^3$) was produced, corresponding to a rate of $\approx 54 \text{ mm yr}^{-1}$. Building on the work of Malamud et al. (2004a), the total volume of landslide material in an event, or period, was used as a measure of the magnitude of the landslide event, or period, defined as the logarithm (base 10) of the total landslide volume produced during the event, or period. From 1937 to 2005, in the study area individual landslide events exhibited a landslide magnitude m_L in the range from 6.0 to 7.3.

The findings are relevant to the determination of landslide hazard and the associated risk in the Collazzone area and in Umbria, and to the evaluation of the evolution of landscapes dominated by mass-

wasting processes in central Italy. We anticipate our work to be a contribution to regional studies of erosion and sediment fluxes, and a starting point for the adoption of a landslide magnitude scale for landslide events.

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